07:30 – 08:30.

## Lecture 3

Fri. 8/03/2024

Announcement!!

Test 1 – 7th April, 2024.

Test 2 – 19th May, 2024.

Derivatives.

- We can now have it that

  - $\frac{d}{dx}(e^x) = e^x$
  - $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$ *
  - $\boxed{\frac{d}{dx}(\ln x) = 1/x}$
  - $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$
  - $\frac{d}{dx}(\ln|u|) = \frac{1}{u}\frac{du}{dx}$ *

__Exercise__
Use first principles to verify.

chain rule where $u = g(x)$

$|-3|$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

short proof

- w.l.o.g, consider $y = \ln|x|$.

Task: To show that $\frac{d}{dx}(\ln|x|) = 1/x$

Case 1: If $x \geq 0$, $y = \ln|x| = \ln x$. $\frac{dy}{dx}$ is obvious. $\frac{dy}{dx} = 1/x$

Case 2: If $x < 0$, $y = \ln|x| = \ln(-x)$

Hence,

$$\frac{dy}{dx} = \frac{d}{dx}(\ln(-x)) = \frac{1}{(-x)}\frac{d(-x)}{dx}$$

$$= \frac{-1}{(-x)}$$

$$= 1/x$$

## Example 1

Find $\frac{dy}{dx}$ if $y = \ln(\overbrace{x^2+1}^{u})$.

Sol$^n$

- Let $\boxed{u = x^2+1}$ Then $\boxed{y = \ln u.}$
- Hence,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{u} \cdot 2x$$

$$= \frac{2x}{u}$$

$$= \frac{2x}{x^2+1} \quad \text{since } u = x^2+1$$

## Exercise

Find $\frac{dy}{dx}$ if

$$y = \log_2 (x^2 + \sin x)$$

Hint:

Use $\frac{d}{dx}(\log_a u) = \frac{1}{u \ln a} \, {}^{du}/_{dx}$

Example 2

Find $\frac{dy}{dx}$ if $y = x^2 e^{\sqrt{x^2+1}}$

Sol$^n$

- To find $\frac{dy}{dx}$, we will use the product rule.

- Thus

$$\frac{dy}{dx} = \frac{d}{dx}\left(x^2 e^{\sqrt{x^2+1}}\right)$$

$$= \frac{d}{dx}(x^2) \cdot e^{\sqrt{x^2+1}} + x^2 \cdot \frac{d}{dx}\left(e^{\sqrt{x^2+1}}\right)$$

$$= 2x e^{\sqrt{x^2+1}} + x^2 e^{\sqrt{x^2+1}} \cdot \frac{1}{\cancel{2}} \boxed{(x^2+1)^{-1/2}} \cancel{2}x$$

$$\frac{1}{(x^2+1)^{1/2}}$$

$$= 2x e^{\sqrt{x^2+1}} + \frac{x^3 e^{\sqrt{x^2+1}}}{\sqrt{x^2+1}}$$

$$= x e^{\sqrt{x^2+1}}\left(2 + x^2/\sqrt{x^2+1}\right)$$

$$= x e^{\sqrt{x^2+1}}\left(\frac{2\sqrt{x^2+1} + x^2}{\sqrt{x^2+1}}\right)$$

$$f(g(x)) = x$$
$$g(f(x)) = x$$

Example 3.

Find $\frac{dy}{dx}$ if $y = 2^x$

$e^{\ln 2^x} = e^{x\ln 2}$

Sol$^n$

Method 1:

- We use the idea that ln and exp are inverses of each other.

$$\ln(e^x) = x$$

$$\boxed{e^{\ln x} = x}$$

$$\ln a^x = x\ln a$$

$$a^x = e^{\ln a^x}$$
$$= e^{x\ln a}$$

Exercise

Find $\frac{dy}{dx}$ if $y = x^x$.

Hence, $y = \boxed{2^x = e^{x\ln 2}}$;

So that

$$\frac{dy}{dx} = \frac{d(2^x)}{dx} = \frac{d}{dx}\left(e^{x\ln 2}\right)$$

Let $u = x\ln 2$

$$\frac{d(e^u)}{dx} = e^u \frac{du}{dx}$$

$$= e^{x\ln 2} \cdot \frac{d}{dx}(x\ln 2)$$

$$= \boxed{e^{x\ln 2}} \ln 2 = 2^x \ln 2.$$

Method 2.

- Use implicit differentiation.

Lecture 4. $\boxed{12:30 - 13:30}$

Find $\frac{dy}{dx}$ if:

1) $y = e^{\tan x}$

2) $y = x^{x^{x}}$

3) $y = \ln(\ln(\ln x))$

4) $y = x^{3x}$

# Integration

- We know that

$$\boxed{\int \frac{1}{x}\,dx = \ln|x| + c}$$

and

$$\boxed{\int e^u\,du = e^u + c}$$ $u = g(x)$

Example

1. Find $\int \frac{\ln x}{x}\,dx$

Let $u = \ln x$. Then $\frac{du}{dx} = \frac{1}{x} \Rightarrow \boxed{du} = \frac{dx}{x}$.

Then $\int \frac{\ln x}{x}\,dx = \int \ln x \cdot \boxed{\frac{dx}{x}} = \int u\,du = \frac{u^2}{2} + c$

$$= \frac{(\ln x)^2}{2} + c$$

2. Find $\int \frac{x}{x^2+1}\,dx$ $\qquad \frac{du}{dx} = 2x$

Sol$^n$

Let $\underline{u = x^2+1}$. Then $du = \boxed{2x\,dx} \Longrightarrow x\,dx = \boxed{\frac{1}{2}\,du}$.

*

Hence,

$$\int \frac{x}{x^2+1}\,dx = \int \frac{1}{x^2+1} \cdot \boxed{x\,dx}$$

$$= \int \frac{1}{u} \cdot \frac{1}{2}\,du = \frac{1}{2}\int \frac{1}{u}\,du$$

$$= \frac{1}{2}\ln|u| + c$$

$$= \frac{1}{2}\ln|x^2+1| + c$$

Since $u = x^2+1$.

Exercise

1. Solve

   1. $\int \frac{dx}{1-3x}$

   $\frac{\sin x}{\cos x}$

   2. $\int \tan x \, dx$ 3. $\int e^x e^{e^x} dx$ ***

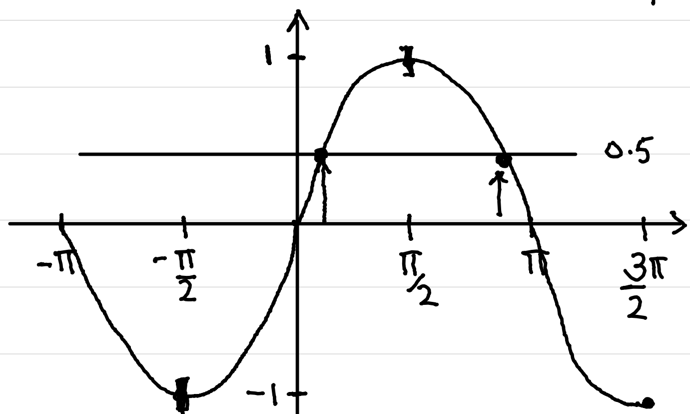
2. Verify that $\int \sec x \, dx = \ln|\sec x + \tan x| + c$

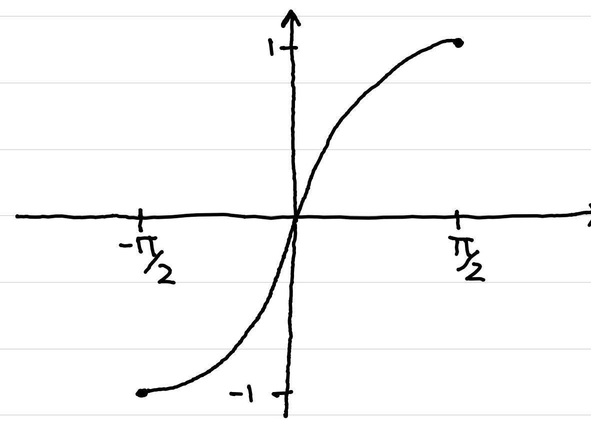
Hints:

- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\sec x = \sec x \frac{(\tan x + \sec x)}{(\tan x + \sec x)}$

# Inverse Trigonometric Functions

- Consider $y = \sin x$ (read as "sine of $x$").
- Its "graph" looks as follows




- ✓ The sine function like other trig functions is not 1-1 hence do not have inverse functions.

✓ One quickest way to check if a function is 1-1 is the horizontal line test. Lecture 4

✓ It fails for trig functions.

✓ We can however <u>restrict</u> the domains of these functions so that they become 1-1.

✓ The inverse function of the restricted sine function exists and is denoted by $\sin^{-1}$ or arcsin.

Recall

$$f^{-1}(x) = y \iff \boxed{f(y) = x}$$

$$\Longrightarrow \boxed{f(f^{-1}(x)) = x}$$

$$\Longleftarrow \boxed{f^{-1}(f(y)) = y}$$

$$f^{-1}(\overbrace{f(y)}^{x}) = y$$

- Hence

$$\sin^{-1}x = y \iff \sin y = x \quad \text{and } -\frac{\pi}{2} \leq y \leq \pi/2$$

- In the same way, we can define arccos, arctan, etc.

$$\cos^{-1}x = y \iff \cos y = x \quad \text{and } 0 \leq \overset{?}{y} \leq \pi$$

$$\tan^{-1}x = y \iff \tan y = x \quad \text{and } -\frac{\pi}{2} < y < \pi/2$$

- The cancelation equations become as

eg

$$\boxed{\begin{array}{ll} \sin^{-1}(\sin x) = x & -\frac{\pi}{2} \leq x \leq \pi/2 \\ \sin(\sin^{-1}(x)) = x & -1 \leq x \leq 1 \end{array}}$$

$\sin^2\theta + \cos^2\theta = 1$

$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Example

Evaluate

a) $\sin^{-1}(1/2) = \pi/6$

b) $\cos(\sin^{-1}(1/2)) = \cos(\pi/6) = \sqrt{3}/2$.

c) $\tan(\underbrace{\sin^{-1}x}_{\theta})$

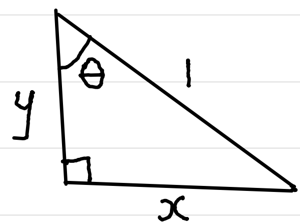
<u>Sol$^n$</u>

Let $\theta = \sin^{-1}x$. Then $\boxed{\sin\theta = x}$ **, $-\frac{\pi}{2} \le \theta \le \pi/2$ and $-1 \le x \le 1$.

Task: Find <u>tan θ</u>.

Method 1.

- We have it that $\sin\theta = \dfrac{x}{1}$ $\left( S = \dfrac{O}{H} \right)$




Hence $y = \sqrt{1-x^2}$

$\therefore \tan\theta = x/\sqrt{1-x^2}$. Thus $\tan(\sin^{-1}x) = x/\sqrt{1-x^2}$

Method 2 (Exercise)